

Classroom Gender Composition: Winners, Losers, and Targeted Assignment

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Central Question

Does the average treatment effect of classroom gender composition conceal persistent winners and losers?

- Growing literature on peer gender effects, but focus on **average** effects
- Null ATE $\neq$ no effect for anyone $\rightarrow$ positive and negative effects may offset
- **Who benefits? Who is harmed? Can we target?**

- **Data**

- Random classroom assignment data of Vietnam in 2011

- **Method**

- Causal Forest (Athey & Wager 2018, 2021) + Policy Learning (Kitagawa & Tetenov 2018)

- **Finding**

- The heterogeneous effects of a higher female share are highly significant
- Beneficiaries losers coexist (30% harmed)
- Policy learning: Causal forest works best for math

- ① **Peer effects:** Document substantial heterogeneity in responses to classroom gender composition, even when average estimates are weak.
- ③ **Causal machine learning:** Use causal forests to recover policy-relevant CATEs in a quasi-random classroom-assignment setting.
- ⑤ **Policy design:** Translate heterogeneity estimates into a selective assignment rule and compare it against econometric and machine-learning benchmarks.

Vietnam Education System (I): Centralized Governance

- **Highly centralized**

- Ministry of Education and Training mandates uniform curricula, pedagogical standards, and core policies nationwide

- **Legal framework**

- 2005 Education Law: equal access and resource allocation
- 2006 Gender Equality Law: prohibits classroom assignment based on ability, gender, or background

- **Strict residential zoning**

- Severely limits active school choice
- Initial school assignment determined by residence, not family selection

Vietnam Education System (II): Quasi-Random Assignment

- **Primary education:** Grades 1–5 (ages 6–10), compulsory and universally coeducational
 - Critical period for cognitive skills, non-cognitive traits, and gender role perceptions
- **Quasi-random class assignment:**
 - Upon Grade 1 entry, students allocated without sorting by achievement, gender, or family background
 - Official guidelines explicitly prohibit using student assessments or parental background as criteria
- **Classes remain intact for 5 years (Grades 1–5):**
 - Students learn all subjects together within a fixed peer group

Data: Young Lives Vietnam

- **Source:** Young Lives survey (U. Oxford)
- **2011 School Survey:** Grade 5
- **Coverage:** 5 provinces, 4 regions

Age	Academic level	YL survey (survey year)
1		Round 1 (2002)
2		
3	Pre-school	
4	Kindergarten	Round 2 (2006)
5	Kindergarten	
6	Grade 1	
7	Grade 2	Round 3 (2009)
8	Grade 3	
9	Grade 4	
<u>10</u>	<u>Grade 5</u>	<u>School Survey (2011)</u>
11	Middle school	Round 4 (2013)
12	Middle school	
13	Middle school	
14	Middle school	Round 5 (2016)
15	High school	
16	High school	
17	High school	Round 6 (2020; COVID)
18	University	
19	University	

Sample Restrictions

- **Sample restrictions**

- Exclude classes formed by ability or age
- Exclude schools with only one Grade 5 class
- Exclude peer groups < 8 students

- **Outcome variable**

- Standardized test score (Math or Vietnamese)

- **Treatment variable**

- Proportion of female students in class c at school site s

- **Control variables**

- Individual: gender, ethnicity, age, and boarding student status
- Family: parental literacy in Vietnamese, the number of children, and household wealth index
- Teacher: gender, age, college degree
- Class: class size

The distribution of treatment variable

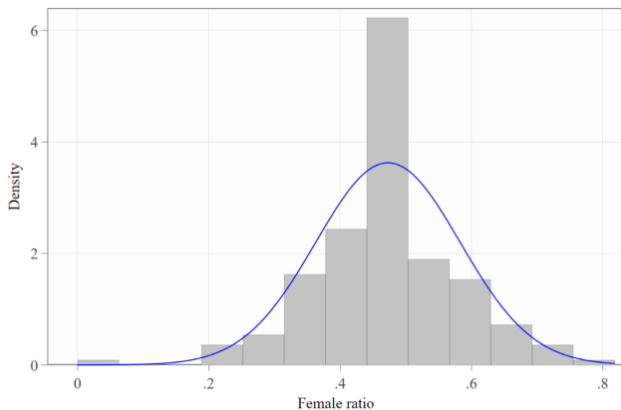


FIGURE 1. Distribution of proportion of females in classrooms

Balance test

	(1) height-for-age z-score	(2) As bad health as other children	(3) Has long term health problems	(4) Log rasch score in the PPVT test	(5) Log rasch score in the CDA test
(Uncorr.) FemaleRate	0.293 (0.532)	0.254 (0.341)	-0.140 (0.088)	0.104 (0.087)	0.059 (0.073)
(Corr.) FemaleRate	0.260 (0.521)	0.225 (0.345)	-0.136 (0.087)	0.113 (0.089)	0.071 (0.073)
Observations	913	912	911	803	881
R-squared	0.146	0.002	0.024	0.354	0.282
	(6) Log food expend. per capita	(7) Log non-food expend. per capita	(8) No choice to decide child's school	(9) Can't help child when sick	(10) Can't help child do well in school
(Uncorr.) FemaleRate	-0.013 (0.150)	0.015 (0.299)	0.315 (0.441)	-0.029 (0.146)	0.141 (0.141)
(Corr.) FemaleRate	-0.024 (0.151)	-0.031 (0.307)	0.301 (0.457)	-0.036 (0.148)	0.091 (0.140)
Observations	914	914	912	914	881
R-squared	0.177	0.391	0.147	0.114	0.067

Balance test

	(11) Ideal ratio of daughters	(12) At what age should child leave full-time education	(13) At what age should child leave household	(14) At what age should child get married	(15) At what age should child have a child
(Uncorr.) FemaleRate	-0.009 (0.043)	-0.161 (1.318)	0.301 (1.257)	0.028 (0.648)	0.201 (0.587)
(Corr.) FemaleRate	-1.133 (2.061)	-0.368 (1.268)	0.432 (1.237)	0.090 (0.623)	0.196 (0.558)
Observations	914	582	690	887	882
R-squared	0.090	0.164	0.328	0.404	0.415

Regression Model

$$Y_{ics} = \alpha + \beta_1 \text{FemaleRate}_{cs} + \beta_2 \text{Female}_i + \theta X_{ics} + \lambda_s + \varepsilon_{ics}$$

- Y_{ics} : standardized test score (Math or Vietnamese) for student i in class c at school s
- FemaleRate_{cs} : proportion of female students in class c at school s
- Female_i : student gender indicator
- X_{ics} : variables at individual, household, teacher and class levels
- λ_s : school fixed effects
- ε_{ics} : error term; standard errors clustered at class level

Result 1: Average Treatment Effect



Table: ATE of Female Ratio on Test Scores

Outcome	Specification	Coef	SE	p	95% CI
Math	School FE Only	0.317	0.180	0.080	[−0.035, 0.670]
Math	FE + Controls	0.237	0.177	<u>0.182</u>	[−0.109, 0.583]
Vietnamese	School FE Only	0.366	0.184	0.049	[0.005, 0.728]
Vietnamese	FE + Controls	0.259	0.162	<u>0.111</u>	[−0.058, 0.576]

Takeaway: ATE is positive but **mostly insignificant**. Had we stopped here, we would conclude gender composition doesn't matter.

Model

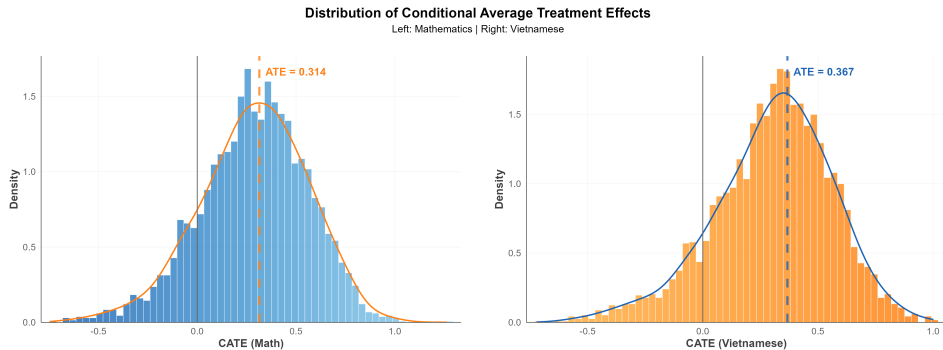
$$Y_i = \mu(X_i, S_i) + \tau(X_i)W_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid X_i, S_i, W_i] = 0$$

- $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$: conditional average treatment effect (CATE)
- School FE removed via DML orthogonalization:

$$\tilde{Y}_i = Y_i - \hat{\mathbb{E}}[Y_i \mid X_i, S_i], \quad \tilde{W}_i = W_i - \hat{\mathbb{E}}[W_i \mid X_i, S_i]$$

- Causal forest on $(\tilde{Y}, \tilde{W}, X)$ with **honesty**, **4000 trees**

Result: CATE Distribution



- CF-ATE: Math = 0.308, Vietnamese = 0.367
- Wide dispersion → many students have **negative** CATE

Result: BLP Heterogeneity Test

- **BLP test** (Chernozhukov et al. 2025, *Econometrica*):

$$\tau(x) = \alpha + \beta \cdot \hat{\tau}(x), \quad \text{test } H_0: \beta = 0$$

Table: Best Linear Projection Test (Chernozhukov et al. 2025)

Outcome	BLP Coef	SE	t-stat	p-value
Math	0.4007	0.0510	7.851	<u>≈ 0</u>
Vietnamese	0.3455	0.0524	6.589	<u>≈ 0</u>

Interpretation

Highly significant heterogeneity ($p < 10^{-10}$).

The CATE varies systematically across students $\rightarrow$ the null ATE masks opposing effects.

Who Benefits vs. Who is Harmed

Table: Proportion of Students by CATE Group

Outcome	Negative (%)	Null (%)	Positive (%)
Math	29.5	37.9	32.6
Vietnamese	29.0	38.6	32.3

Key Finding

- ~30% of students experience **negative** effects from increased female ratio.
- ~38% experience null effects.
- ~32% experience positive effects.
- The insignificant ATE is a **precarious balance** of opposing forces.

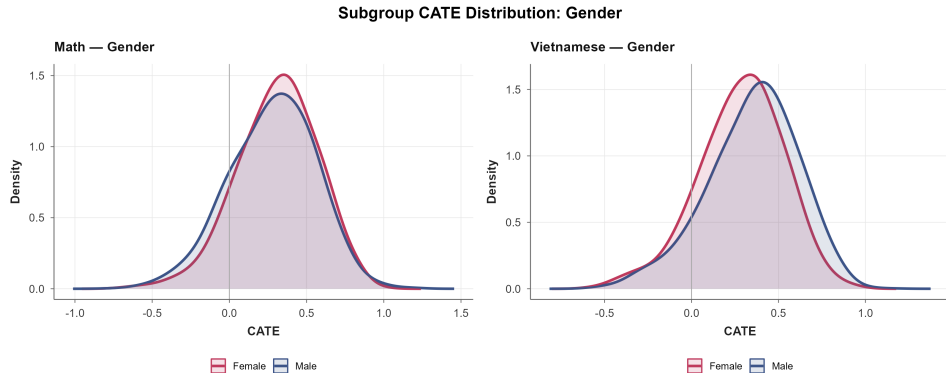
Who Benefits vs. Who is Harmed

- To better characterize which students benefit from classroom gender composition, which remain unaffected, and which are harmed, we apply a simple k means clustering algorithm.

Table: Characteristics of Most vs. Least Affected (Top 20% CATE)

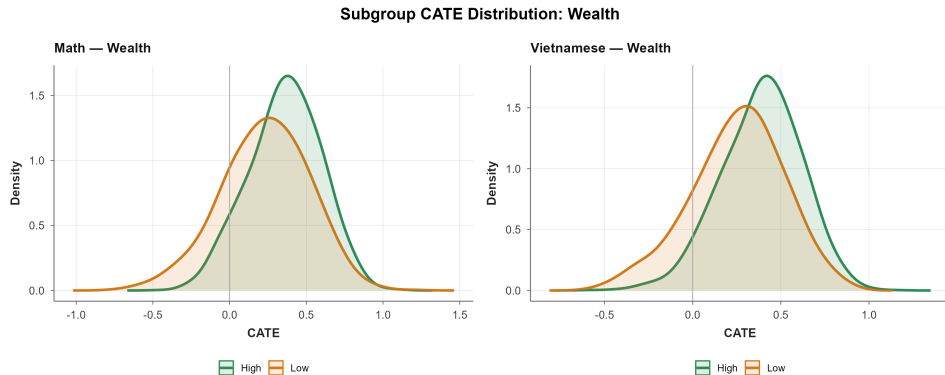
Outcome	Variable	<u>Most</u>	<u>Least</u>	Diff.	<i>p</i>
Math	Class Size	26.3	30.3	-4.09	< 0.001
Math	Wealth Index	0.631	0.486	+0.145	< 0.001
Math	Female	0.507	0.404	+0.102	< 0.001
Math	Minority	0.040	0.186	-0.146	< 0.001
Vietnamese	Wealth Index	0.626	0.505	+0.122	< 0.001
Vietnamese	Teacher Exp.	16.4	18.7	-2.32	< 0.001
Vietnamese	Female	0.362	0.547	-0.186	< 0.001
Vietnamese	Minority	0.070	0.118	-0.048	= 0.001

Mechanism 1: Gender — Stereotype Threat & Catch-Up



- **Math:** Female CATE = 0.310 > Male = 0.279 ($\Delta = 0.031$, $p < 0.001$) → **stereotype threat mitigation**
- **Vietnamese:** Male CATE = 0.351 > Female = 0.283 ($\Delta = -0.068$, $p < 0.001$) → **catch-up effect**
- **Complementarity:** those weaker in a domain gain more from opposite-gender peers

Mechanism 2: Wealth — Resource Amplification



- **Math:** High-wealth CATE = 0.360 vs. Low = 0.231 ($\Delta = 0.129$, $p < 0.001$)
- **Vietnamese:** High-wealth CATE = 0.389 vs. Low = 0.252 ($\Delta = 0.136$, $p < 0.001$)
- **Both outcomes:** wealthy students benefit more from female peers

Policy Learning Framework

Setup

- **Train/test split:** 70/30, stratified by wealth tercile $\times$ gender $\times$ ethnicity
- **School FE:** Demeaned separately for train and test
- **Treatment:** ratio_female (continuous) — no binarization
- **Evaluation:** Custom RATE function for continuous treatment

Methods Compared

- ① **Causal Forest:** Nonparametric CATE estimation
- ② **OLS:** Linear model with $W \times X$ interactions
- ③ **LASSO:** Penalized regression with interactions
- ④ **Random Forest (R-learner):** Nonparametric alternative
- ⑤ **Policy Tree:** DR-score-based assignment (Athey & Wager 2021)

Custom RATE for Continuous Treatment

Why Custom?

`grf::rank_average_treatment_effect()` only supports binary W .
We implement custom AUTOC/QINI for continuous treatment.

Definition

$$\begin{aligned}\text{TOC}(q) &= \mathbb{E}[\hat{\tau}_{\text{gt}}(X) \mid S(X) \geq F_S^{-1}(1 - q)] - \mathbb{E}[\hat{\tau}_{\text{gt}}(X)] \\ \text{AUTOC} &= \int_0^1 \text{TOC}(q) dq, \quad \text{QINI} = \frac{\int_0^1 (1 - q) \cdot \text{TOC}(q) dq}{\int_0^1 (1 - q) dq}\end{aligned}$$

where $\hat{\tau}_{\text{gt}}$ = test-set CF predictions (ground truth), $S(X)$ = priority score.
SE via bootstrap ($B = 500$).

References: Athey & Wager (2021, *Econometrica*); Yadlowsky et al. (2025, *JASA*).

Result: Policy Comparison (RATE)

Table: AUTO C and QINI by Method

Outcome	Method	AUTO C	SE	QINI	SE
Math	OLS	0.0646	0.0073	0.0997	0.0110
Math	Causal Forest	0.0800	0.0070	0.1048	0.0111
Math	Random Forest	0.0582	0.0076	0.0825	0.0106
Math	LASSO	0.0572	0.0074	0.0896	0.0109
Vietnamese	OLS	0.0768	0.0133	0.0959	0.0204
Vietnamese	Causal Forest	0.0071	0.0138	0.0016	0.0194
Vietnamese	Random Forest	0.0035	0.0117	0.0106	0.0189
Vietnamese	LASSO	0.0368	0.0146	0.0431	0.0218

Why Does OLS Win for Vietnamese? — Overoptimism Analysis

Table: Predicted vs. Actual CATE: Overoptimism

Outcome	Target	Method	Predicted	Actual	Over-opt.	Over-opt. Ratio (%)
Vietnamese	10%	CF	0.817	0.215	0.602	279.5
Vietnamese	10%	OLS	1.342	0.351	0.992	282.8
Vietnamese	10%	RF	1.279	0.269	1.010	375.9
Math	10%	CF	0.443	0.961	-0.517	<u>-53.8</u>
Math	10%	OLS	1.865	1.023	0.842	<u>82.4</u>

Fairness of Targeting (I): Gender & Ethnicity

Table: Demographic Composition of Targeted Groups

Outcome	Target	Method	Female (%)	Minority (%)
Vietnamese	10%	CF	<u>73.0</u>	<u>7.0</u>
Vietnamese	10%	OLS	45.2	6.1
Vietnamese	10%	RF	57.4	15.7
Math	10%	CF	<u>56.5</u>	<u>2.6</u>
Math	10%	OLS	41.7	5.2

Fairness of Targeting (II): Socioeconomic Status

Table: Socioeconomic Composition of Targeted Groups (10%)

Outcome	Target	Method	Low Wealth (%)	Low Parent Lit (%)	Boarder (%)	Older (%)
Vietnamese	10%	CF	<u>50.4</u>	6.1	2.6	3.5
Vietnamese	10%	OLS	40.9	7.8	0.9	7.0
Vietnamese	10%	RF	58.3	15.7	1.7	7.0
Math	10%	CF	<u>18.3</u>	3.5	5.2	0.9
Math	10%	OLS	43.5	8.7	18.3	7.0

- **What we can confidently say**

- Heterogeneity exists (BLP $p < 10^{-10}$)
- ~30% of students are harmed
- CF outperforms OLS for Math (AUTOOC 0.080 vs. 0.065)

- **Key limitations**

- No spillover modeling (individualistic intervention assumption)
- “Ground truth” CATE is estimated, not observed
- Suggestive mechanism, not causal inference
- Carrell et al. (2013) Lucas critique: endogenous peer response may undermine policy

Thank you!
Your comments are welcome.